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Bidirectional transformer with knowledge graph for video captioning

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Abstract

Models based on transformer architecture have risen to prominence for video captioning. However, most models are only to improve either the encoder or the decoder, because when we improve the encoder and decoder simultaneously, the shortcomings of either side may be amplified. Based on the transformer architecture, we connect a bidirectional decoder and an encoder that integrates fine-grained spatio-temporal features, objects, and relationships between the objects in the video. Experiments show that improvements in the encoder amplify the information leakage of the bidirectional decoder and further produce a worse result. To tackle this problem, we generate pseudo reverse captions and propose a Bidirectional Transformer with Knowledge Graph (BTKG), which integrates the outputs of two encoders into the forward and backward decoders of the bidirectional decoder, respectively. In addition, we make fine-grained improvements on the interior of the different encoders according to four modal features of the video. Experiments on two mainstream benchmark datasets, i.e., MSVD and MSR-VTT, demonstrate the effectiveness of BTKG, which achieves state-of-the-art performance in significant metrics. Moreover, the sentences generated by BTKG contain scene words and modifiers, that are more in line with human language habits. Codes are available on https://github.com/nickchen121/BTKG.

Keywords Video captioning · Bidirectional transformer · Knowledge graph · Multimodal of video

1 Introduction

The task of video captioning aims to understand the scenes in a video and describe them with plausible sentences. It can be widely used in video retrieval, video recommendation, disabled supporting, and scene understanding. With the rapid development of deep learning, the encoder-decoder-based neural caption methods [1–4] have risen to prominence for video captioning. Especially with the emergence of Transformer [5], models based on transformer

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architecture [6–11] have been successful in advancing the state-of-the-art. However, the above models only improve either the encoder or the decoder, e.g., by integrating objects in the video and relationships between the objects into the encoder [7], by fusing audio, image, and motion features of the video into the encoder [9], and by employing a bidirectional decoder [8]. Therefore, the motivation for simultaneously improving the encoder and decoder of Transformer is natural, e.g., the decoder generates caption with higher accuracy through a bidirectional decoder while fusing more modal features of the video into the encoder.

As shown in MEV of Fig.1, Zhang et al. [7] used the state-of-the-art 2-D CNN and 3-D CNN pre-trained on a large dataset to extract visual spatio-temporal features [12]. They also acquired the relationships between objects extracted based on the object detector Mask R-CNN [13] and the knowledge graph model TransE [14]. Existing unidirectional decoders generate the target language sequence token-by-token from left to right, and they cannot make full use of the target-side future contexts, which can be produced in a right-to-left decoding direction [15]. Therefore, Wang et al. [8] proposed a bidirectional decoder to address the above problem. As shown in BDD of Fig. 1, the bidirectional decoder is composed of a Forward Decoder (FD) and a Backward Decoder (BD), because the forward decoder accept the reverse captions generated by the backward decoder, the forward decoder can considers the previously generated outputs and the following words of the ground-truth caption from the backward decoder.

Nevertheless, the bidirectional decoder exists a problem of information leakage [16] . The leakage is that in the training stage, the forward decoder indirectly utilizes the reverse

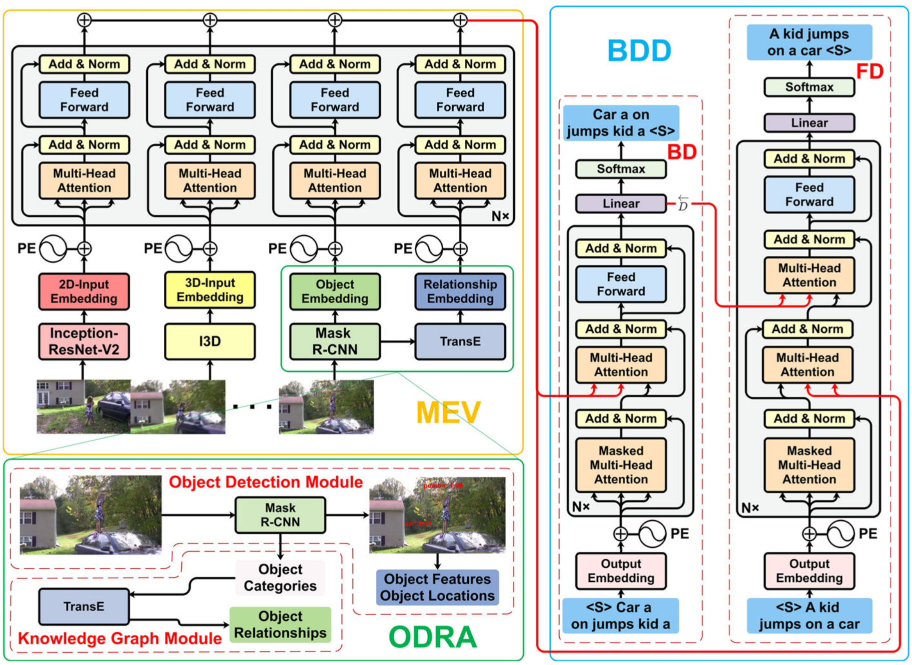

Fig. 1 The figure illustrates a direct connection between a Multimodal Encoder of Video (MEV) and a Bidirectional Decoder (BDD) based on the transformer architecture, where PE is the positional encoding. ODRA is a combination between a Knowledge Graph Module and an Object Detection Module, in which Mask R-CNN and TransE represent the object detector and the knowledge graph model, respectively

ground-truth caption, which is the input from the backward decoder. On the contrary, the forward decoder employs the reverse caption generated by the backward decoder in the testing stage. Therefore, if the backward decoder generates higher quality reverse caption, it is equivalent to that the forward decoder uses caption closer to the reverse ground-truth caption in the testing stage, i.e., as long as the more outstanding the performance of the backward decoder is, the less negligible negative impact of the leakage on the bidirectional decoder is. As shown in ODRA of Fig. 1, we acquire the objects ('car', 'people') from an object detector Mask R-CNN. Since objects are composed of a series of isolated entities and no reverse order concept between contexts, it is not very sensible for us to reverse them. The same goes for the relationships obtained from a knowledge graph model TransE. Therefore, if we integrate the objects and the relationships into the backward decoder, the quality of generating reverse caption may be reduced and further expand the leakage. As shown in Fig. 1, when we directly connect a bidirectional decoder and an encoder that fuses the objects and the relationships, experiments show that it does amplify the leakage.

Therefore, we propose a Bidirectional Transformer with Knowledge Graph (BTKG) to mitigate the leakage. In Fig. 2, we respectively assign one encoder to each of the forward and backward decoders of the bidirectional decoder, which is different from the conventional model with only one encoder and decoder. Since the encoder integrating the objects and the relationships can reduce the generation quality of reverse caption, as shown in STE of Fig. 2, we remove the objects and relationships for the encoder connected to the backward decoder. In addition, we generate pseudo reverse captions to reduce the leakage by exploiting the characteristic of having multiple various captions for a video [8].

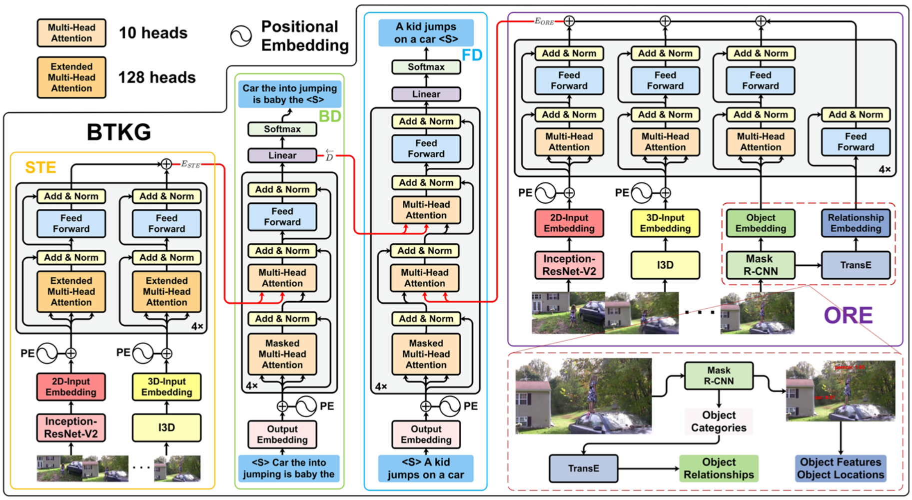

Fig. 2 The figure illustrates our proposed BTKG based on the transformer architecture in detail, which consists of Spatio-Temporal Encoder (STE), Objects and Relationships Encoder (ORE), Backward Decoder (BD), and Forward Decoder (FD). In training, given a video, fine-grained spatio-temporal features of the video E_{STE} is obtained through the STE. Then we input E_{STE} into the BD to acquire the reverse predicted caption Y_{BD} and the context $\overleftarrow{D}$ of the Y_{BD}. At the same time, for acquiring the video encoding E_{ORE}, coarse-grained spatio-temporal features of the video, the objects obtained through an object detector Mask R-CNN, and the relationships between the objects extracted based on a knowledge graph model TansE are jointly encoded by the ORE. Finally, the E_{ORE} is integrated into the FD together with the $\overleftarrow{D}$ to obtain the forward predicted caption Y_{FD}

Moreover, different from the conventional encoder of the Transformer for sequential texts, the inputs of our encoders are frames, objects, and relationships in the video. Therefore, as shown in Fig. 2, we employ different encoders according to the various modals of the video. Because frames in the video have tighter correlations than words in a text, we first capture the correlations between smaller video clips than frames through an extended and multi-head attention layer in STE. Secondly, each relationship is independent, so there is no need to employ an attention layer for capturing the correlations between them. Finally, since neither the objects nor the relationships are contextually ordered, neither utilizes a positional encoding.

The main contributions are summarized in the following four aspects:

- To utilize the bidirectional decoder when integrating objects in a video and relationships between the objects into the encoder, we propose a Bidirectional Transformer with Knowledge Graph (BTKG). It allows two different encoders to connect the forward and backward decoders, respectively.
- The inputs of our encoders are modal features of the video, which is different from the Transformer for sequential text. Therefore, we improve the encoders according to the different modal features.
- To mitigate the information leakage of bidirectional decoder, we generate pseudo reverse captions. In addition, the extended and multi-head attention layer indirectly improves the performance of the encoder and the backward decoder and further alleviates the leakage.
- We evaluate BTKG on two datasets, MSVD and MSR-VTT. In particular, our BTKG achieves state-of-the-art performance and outperforms the runner-up methods by a large margin in CIDEr-D, which is specially designed for captioning tasks. Moreover, the sentences generated by BTKG contain scene words and modifiers, that are more in line with human language habits.

The rest of this paper is organized as follows. We first discuss related work on video captioning in Section 2. Secondly, the Bidirectional Transformer with Knowledge Graph (BTKG) is detailed in Section 3. Then, we present the experimental results and analyses in Sections 4 and 5, respectively. Finally, we conclude the paper in Section 6.

2 Related work

In this section, we describe the related work from 1) Video Captioning, 2) Multimodal Encoding, 3) Captions Generation, and 4) Summary.

2.1 Video captioning

Recently, captioning a short video in natural language has been challenging for machines. With the rapid development in deep learning, Donahue et al. are the first to adopt a deep neural network to solve the video captioning problem [1]. Later, many video captioning methods based on encoder-decoder architecture rose to prominence [1–4]. These methods encode the video by a Convolutional Neural Network (CNN) [17] and employ a Long Short-Term Memory (LSTM) [18] to generate video captions. One of the first works that utilize an encoder-decoder framework is Venugopalan et al., where captions are generated by LSTM and visual features extracted by CNN [2]. With the emergence of the attention mechanism and Transformer [5], Zhou et al. also propose a Transformer method to utilize an attention mechanism to generate coherent captions [6]. Zhong et al. propose a three-layer hierarchical attention network based on a bidirectional decoding transformer that enhances multimodal features [19].

2.2 Multimodal extraction

In order to exploit the temporal structure of the video, the utilization of multi-modalities in video encoding also attracted great attention. For instance, Song et al. extend the importance of implicitly distinguishing visual and non-visual words through a hierarchical attention mechanism [20]. In particular, Aafaq et al. leverage the state-of-the-art 2-D CNN and 3-D CNN (C3D [21]) pre-trained on a large dataset to extract visual spatio-temporal features [12]. With the success of object detection in Computer Vision, the bottom-up attention algorithm applies object detection to extract regional features, significantly improving the video captioning performance [22]. Yan et al. extract the local features of maintaining more accurate object information via object detector [23]. Zhang et al. respectively utilize object detection and knowledge graph to extract objects and relationships between the objects in the video and then fuse them with spatio-temporal features of the video to refine the fine-grained actions between the objects [7]. Zhou et al. proposed a novel MATNet to introduce a new way to learn rich spatiotemporal object representations with an interleaved encoder, which encourages knowledge propagation from motion to appearance in a hierarchical manner [24].

2.3 Captions generation

Recently, many studies have also made some achievements in captions generation. Liu et al. tackled the captioning problem of the controllable video by injecting robust control signals into the selected objects [25]. For revising word and grammar errors, Xu et al. propose DRPN to refine the caption candidates [26]. In order to capture more visually grounded details from videos, Yang et al. designed a special decoding algorithm to realize coarse-to-fine rather than the caption procedure word-by-word [27]. The decoder of the Transformer is an autoregressive decoder, which only considers the currently generated words. Wang et al. propose a bidirectional decoding model for generating captions by using the context [8].

2.4 Summary

As mentioned above, the above methods almost only improve the encoder or the decoder. Experiments show that direct superposition of the encoder and decoder based on [7] and [8] may produce a worse results. Inspired by these works, we propose BTKG based on transformer architecture to effectively fuse the improved encoder and decoder. The inputs of our encoders are frames, objects, and relationships in the video, which is different from the Transformer for sequential text. Based on the difference, we improve the encoders according to the various modal features of the video.

3 Approach

In this section, we first briefly introduce the used basic module of our approach and then describe the details of the approach.

3.1 Basic module

The basic module are introduced from a transformer architecture and the used multimodal features extraction and captions generation.

3.1.1 Transformer architecture

Transformer architecture [5] consists of an encoder and a decoder, which are composed of a stack of identical layers. Each layer contains sublayers constructed by Multi-Head Attention (MHA) and Feed-Forward Network (FFN). At the same time, there is a residual connection around each of the two sub-layers, followed by layer normalization.

The MHA sublayer allows the model to jointly attend to information from different representation subspaces at different positions and consists of h identical heads. Each head$_i$ utilizes $Q_i = QW_i^Q, K_i = KW_i^K, V_i = VW_i^V$ as its input. The MHA is calculated as follows:

$$\begin{aligned} &\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O, \\ &\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \end{aligned} \tag{1}$$

where W_i^Q, W_i^K, W_i^V are learned parameters, and W^O is a learned projection matrix.

The attention of (1) is called "Scaled Dot-Product Attention". It can be described as mapping a query and a set of key-value pairs to an output:

$$\text{Attention}(Q_i, K_i, V_i) = \text{softmax}(\frac{Q_i K_i^T}{\sqrt{d_k}})V_i, \tag{2}$$

where $E_i = \frac{Q_i K_i^T}{\sqrt{d_k}}$ represents the $N \times N$ attention score matrix E_i, whose element e_i^{mn} is the attention score between the m-th and n-th token in i-th subspace.

The FFN sublayer consists of two linear transformations with a ReLU activation in between:

$$FFN(X) = max(0, XW_1 + b1)W_2 + b_2, \tag{3}$$

where $W_1 \in \mathbb{R}^{d \times d_f}$, $W_2 \in \mathbb{R}^{d_f \times d}$, $b_1 \in \mathbb{R}^{d_f}$, $b_2 \in \mathbb{R}^d$ are learned parameters, and $d = d_k \times h$.

3.1.2 Multimodal features extraction and captions generation

Image and motion features have been widely used for multimodal features extraction in video. Image features are generally used to express the colors and the shapes in the image, and motion features are essential for capturing spatio-temporal motion dynamics in the video. We employ the Inception-ResNet-V2 [28] pre-trained on the ImageNet [17] to extract the image features and utilize the I3D [29] pre-trained on the Kinetics [30] to extract the motion features. In addition, objects in the video have a dynamic behavior compared to those in the picture, and there are relation links between them. We first leverage the object detector Mask R-CNN [13] to extract the semantic objects (such as people, buildings, cars) and then acquire relationships between the objects through the knowledge graph model TransE [14], which is based on the knowledge representation learning framework OpenKE [31]. For captions generation, our approach adapts from the bidirectional decoder [8], which consists of a backward decoder and a forward decoder.

3.2 Bidirectional transformer with knowledge graph (BTKG)

As stated above, our Bidirectional Transformer with Knowledge Graph (BTKG) combines a bidirectional decoder proposed by Wang [8] with an encoder proposed by Zhang [7] that integrates objects in the video and relationships between the objects. As shown in Fig. 2, BTKG consists of four components: given a video, **Spatio-Temporal Encoder (STE)** obtains fine-grained video spatio-temporal encoding E_{STE} of the video exploiting an extended and multi-head attention layer, **Objects and Relationships Encoder (ORE)** first extracts objects in the video through object detector Mask R-CNN and acquires the relationships between the objects through the knowledge graph model TransE, then jointly encode the objects, the relationships, and the coarse-grained spatio-temporal features to obtain E_{ORE}, **Backward Decoder (BD)** generates reverse predicted caption Y_{BD} and the context $\overleftarrow{D}$ of the Y_{BD}, and **Forward Decoder (FD)** employs E_{ORE} and $\overleftarrow{D}$ to perform forward decoding for acquiring the forward predicted caption Y_{FD}. The details of each component are described in the following subsections.

3.2.1 Spatio-temporal encoder (STE)

The Spatio-Temporal Encoder (STE) is expected to encode fine-grained video spatio-temporal features. Given a sequence of video frames $X = \{x_1, \cdots, x_L\}$ with length L, we first leverage the Inception-ResNet-V2 pre-trained on the ImageNet to extract image features $F_I = \{f_1, \cdots, f_L\}$ where $f_i \in \mathbb{R}^{d_I}$, and utilize the I3D pre-trained on the Kinetics to extract motion features $F_M = \{v_1, \cdots, v_N\}$ where $v_i \in \mathbb{R}^{d_M}$ and $N = L/64$. The dimensions of image and motion features are represented by d_I and d_M, respectively. Then we make linear changes for the image and motion features to map their dimensions as d_{model}, the calculation formula is as:

$$f'_L = w_L f_T + b_L,\ b_L \in \mathbb{R}^{d_{model}}, \tag{4}$$

$$v'_N = w_N v_N + b_N,\ b_N \in \mathbb{R}^{d_{model}}, \tag{5}$$

where $w_L \in \mathbb{R}^{d_{model} \times d_I}$ and $w_N \in \mathbb{R}^{d_{model} \times d_M}$. We can obtain the $F'_I = [f'_1, \cdots, f'_i],\ f'_L \in \mathbb{R}^{d_{model}}$ and $F'_M = [v'_1, \cdots, v'_N],\ v'_i \in \mathbb{R}^{d_{model}}$ through the linear changes. Next, we send F'_I and F'_M into different encoders as input to acquire $E_I \in \mathbb{R}^{L \times d_{model}}$ and $E_M \in \mathbb{R}^{N \times d_{model}}$. Finally, we connect the encoded image and motion features to acquire the output $E_{STE} \in \mathbb{R}^{L \times d_{model}}$ of STE as:

$$E_{STE} = E_I \oplus E_M, \tag{6}$$

where $\oplus$ denotes the element-wise addition.

Extended multi-head attention The video is artificially cut into frames according to time, and the multi-head attention (Vaswani et al. utilize 8 heads) allows the model to jointly attend to information from different representations subspaces at different positions [5]. In STE, when we employ the extended and multi-head attention layer (Our model uses 128 heads, that is, $h = 128$ in (1)), BTKG can capture the correlation of different positions in the time clips smaller than frames.

3.2.2 Objects and relationships encoder (ORE)

In order to provide finer-grained actions between objects, we obtain relationships between the objects through a knowledge graph model. As shown in ODRA of Fig. 1, we utilize the object detector Mask R-CNN to obtain objects and choose TransE based on the knowledge representation learning framework OpenKE to acquire relationships between the objects.

Object detection We extract the location and eigenvector of objects from Mask R-CNN pre-trained based on COCO [32] dataset and obtain categories of the objects from 50 common categories. Specifically, we first input a video V into the Feature Pyramid Network [33] to obtain feature maps of the video. Then we input them into the Region Proposal Network [34] to filter out a series of Regions of Interest (ROI). Finally, because it extracts the different sizes of the ROI, we employ an ROI pooling layer to acquire the same size of ROI, namely Object Regions (OR). Where the position coordinates of the OR, the eigenvector, and the categories are expressed as $R_l = [l_1, \cdots, l_k]$, $R_v = [v_1, \cdots, v_k]$ and $R_o = [o_1, \cdots, o_k]$, respectively, where k represents the k-th detected object, $l_i = [\frac{x_i}{w_f}, \frac{y_i}{h_f}, \frac{w_i}{w_f}, \frac{h_i}{h_f}]$ provides the center coordinates $(\frac{x_i}{w_f}, \frac{y_i}{h_f})$, width $\frac{w_i}{w_f}$ and height $\frac{h_i}{h_f}$ of the i-th object area after normalization according to the size of the video frame with width w_f and height h_f, and v_i represents the eigenvector of the i-th object. Last we splice R_L and R_V as the objects feature, indicating F_o :

$$F_o = concat_{i=1,\ldots,k}(R_l R_v). \tag{7}$$

Relationships acquisition based on knowledge graph We first leverage the categories R_o as the input of TransE to predict relationships $R_r = [r_1, \cdots, r_{\frac{(k-1)k}{2}}]$ between the objects, and then utilize Word2Vec [35] to encode R_o and R_r into word vector. The encoding process is as:

$$o_i^{'} = o_i W, \tag{8}$$

$$r_i^{'} = r_i W, \tag{9}$$

where $r_i = [0, \cdots, 1, \cdots, 0]$, $r_i \in \mathbb{R}^{\frac{k^2}{2}}$ represents the one-hot vector, and the embedding matrix W is pre-trained by Word2Vec. Finally, the final relationships feature synthesized by $R_o^{'} = [o_1^{'}, \cdots, o_k^{'}]$ and $R_r^{'} = [r_1^{'}, \cdots, r_{\frac{(k-1)k}{2}}^{'}]$ is as:

$$F_r = [R_o^{'};\ R_r^{'}], \tag{10}$$

where $[\cdot; \cdot]$ denotes concatenation.

Objects and relationships encoding The Objects and Relationships Encoder (ORE) first employs the same method as STE to obtain image features $F_I^{'}$ and motion features $F_M^{'}$. The difference is that ORE utilizes a 10-head attention layer to encode $F_I^{'}$ and $F_M^{'}$ for getting the image encoding E_I and the motion encoding E_M. Next, we encode F_o to acquire the objects encoding E_o, where we do not utilize the positional encoding because they are not contextually ordered. Since the feature of relationships $F_r = [o_1^{'}, \cdots, o_k^{'}; r_1^{'}, \cdots, r_{\frac{(k-1)k}{2}}^{'}]$ like the objects F_o, positional encoding is also not used in the relational encoder. However, there is irrelevant between $r_i^{'}$ and $r_j^{'}$ in F_r, where i and $j \in [1, \frac{(k-1)k}{2}]$, so we employ the encoder without an attention layer to encode F_r for acquiring the relationships encoding E_r.

Finally, we combine the above four encodings as the output E_{ORE} of ORE:

$$E_{ORE} = E_I \oplus E_M \oplus E_o \oplus E_r. \tag{11}$$

3.2.3 Backward decoder (BD)

The Backward Decoder (BD) generates a reverse caption from right to left. Because the objects and relationships have no concept of reverse order, which will reduce the performance of the BD, we cannot integrate the output E_{ORE} of the Objects and Relationships Encoder. Therefore, we integrate the output E_{STE} into the BD. When the predicted word is the end marker $< \mathrm{S} >$, the prediction of the reverse caption ends. We express it as $\overleftarrow{C} = [s_1, \cdots, s_L, < \mathrm{S} >]$. As shown in BD of Fig. 2, we also acquire the context $\overleftarrow{D}$ of the reverse caption as:

$$\overleftarrow{D} = \overleftarrow{D_L}, \tag{12}$$

where $\overleftarrow{D_L}$ is the hidden state obtained from the Linear layer when the BD generates the last word s_L.

3.2.4 Forward decoder (FD)

The Forward Decoder (FD) predicts a word from left to right by integrating the context of the reverse caption generated by the BD, i.e., compared with the BD, the FD adds a cross multi-head attention layer to integrate the context $\overleftarrow{D}$, so that the forward decoder can take into account the context of the ground-truth caption every time it predicts the next word. At the same time, because the objects and the relationships enhance the feature representation of the video and cannot damage the performance of the FD, so we integrate the output E_{ORE} into the FD. When the predicted word is the end marker $< \mathrm{S} >$, the prediction of the forward caption is completed, which is expressed as $\overrightarrow{C} = [s_1, \cdots, s_T, < \mathrm{S} >]$.

3.3 Training

During the training of BTKG, one of the most crucial points is the introduction of multimodal encoding with different combinations and bidirectional decoding. We utilize a two-stage method to generate video captions in bidirectional encoding. Specifically, we first obtain the pseudo reverse caption through the backward decoder and then integrate it into the forward decoder to generate a forward caption. For bidirectional decoding, we introduce the typical cross-entropy losses $\mathcal{L}_{bd}$ and $\mathcal{L}_{fd}$ for the backward and forward decoders, respectively. Given a video V, its ground-truth caption $\overrightarrow{Y} = [y_1, \cdots, y_L]$ of length L, and the reverse caption $\overleftarrow{Y} = [y_1, \cdots, y_T]$ of length T from a training dataset $\mathcal{D}$, the loss formula of bidirectional decoder $\mathcal{L}$ is as:

$$\mathcal{L} = (1 - \lambda)\mathcal{L}_{bd} + \lambda\mathcal{L}_{fd}, \tag{13}$$

where $\lambda \in [0, 1]$ is a hyper-parameter used to balance the preferences between two decoders. Next, we will explain the pseudo reverse captions and these two losses in detail.

3.3.1 Pseudo reverse captions

Because we generate pseudo reverse captions to mitigate the information leakage of bidirectional decoder, the lengths of the $\overleftarrow{Y}$ and the $\overrightarrow{Y}$ are inconsistent. Specifically, we first reverse all the ground-truth captions of a video to obtain the corresponding reverse captions of the same length. Then we randomly disorder these reverse captions to obtain the pseudo reverse captions corresponding to the forward captions, which leads to the final reverse caption is not the inversion of the ground-truth caption.

3.3.2 Backward decoder loss

Backward decoder loss is defined as the negative log-likelihood to generate the reverse caption:

$$\begin{aligned}\mathcal{L}_{bd} = &\sum_{(V,\overleftarrow{Y})\in\mathcal{D}} \sum_{t} (-\log\ p(y_t|V, y_1, \cdots, y_{t-1}; \\ &\theta_{ste}; \theta_{bd})),\end{aligned} \tag{14}$$

where y_t is the word being predicted, θ_{ste} and θ_{bd} are the learnable parameters of Spatio-Temporal Encoder and Backward Decoder, respectively.

3.3.3 Forward decoder loss

The traditional transformer decoder masks its input so that the decoder cannot refer to the following words of the ground-truth caption when predicting a word of the caption. Therefore, we utilize a bidirectional decoder, which aims to obtain the following words from $\overleftarrow{D}$ generated by the backward decoder. Then $\overleftarrow{D}$ is integrated into the word generation of the forward caption when the FD generates the current word. The cross-entropy loss of the forward decoder is defined similarly as (14):

$$\begin{aligned}\mathcal{L}_{fd} = &\sum_{(V,\overrightarrow{Y})\in\mathcal{D}} \sum_{l} (-\log\ p(y_l|V, y_1, \cdots, y_{l-1}; \\ &\theta_{ste}; \theta_{bd}; \theta_{ore}; \theta_{fd}),\end{aligned} \tag{15}$$

where θ_{ore} and θ_{fd} are learnable parameters of Objects and Relationships Encoder and Forward Decoder, respectively.

4 Experiments

In this section, We first describe four experimental settings: Datasets, Feature Extraction, Evaluation Metrics, and Parameter Settings, then introduce the Performance Comparison of our method.

4.1 Experimental settings

4.1.1 Datasets

Various experiments are conducted using the two most popular benchmark datasets to demonstrate the effectiveness of our proposed BTKG, which are Microsoft Research video captioning Corpus (MSVD) [36], and Microsoft Research video to text (MSR-VTT) [37].

MSVD It contains 1,970 YouTube short video clips in 10 seconds to 25 seconds and roughly 80,000 English sentences. Each video clip depicts a single activity and has about 40 English captions. We follow the split settings in prior works [38, 39], resulting in 1,200, 100, and 670 videos for the training, validation, and test set, respectively.

MSR-VTT It consists of 10,000 video clips, each of which has 20 captions and a category tag annotated by 1327 workers from Amazon Mechanical Turk. Like [40], there are 6,513, 497, and 2,990 videos for the training, validation, and test set, respectively.

4.1.2 Feature extraction

We sample the videos of MSVD and MSR-VTT at 5 and 3 frames per second, respectively. Then we input the sampled results into the Inception-ResNet-V2 to extract the image features and re-sample the videos of MSVD and MSR-VTT at the rate of 25 and 15 frames per second, respectively. Finally, we input the sampled results into the I3D to obtain motion features extracted from 64 overlapping continuous frames, and the interval for extracting the motion features is set to 5 frames. The dimensions of image and motion features are 2048-D and 1024-D, respectively. Moreover, we also employed GLOVE [41] vectors of the auxiliary video category labels to facilitate feature encoding. We extract 50 frames and 60 frames from the visual features in MSVD and MSR-VTT at equal intervals, respectively. In this paper, both features are projected to $d_h = 512$.

4.1.3 Evaluation metrics

We use standard automatic evaluation metrics to test the performance of the model, including BLEU [42], METEOR [43], CIDEr-D [44], and ROUGE-L [45]. BLEU and METEOR are initially designed for machine translation evaluation and are commonly used to quantify the quality of a machine-generated text. CIDEr-D is a recently introduced evaluation metric for image caption tasks, designed explicitly for evaluation caption systems, while ROUGE-L was used to evaluate the extracted text summarization proposed in 2004.

4.1.4 Parameter settings

For obtaining the objects and the relationships between the objects, the confidence of Mask R-CNN is 0.7, and its minimum detection size is 224×224. At the same time, we utilize the 613 non-repeated triples as the dataset of TransE and then employ the 300-dimensional GLOVE vectors to represent the relationships. We adopt 4 layers, 640 word embedding dimensions, 512 model dimensions, 2048 hidden dimensions, and 10 attention heads per layer for the above encoders and decoders. However, for the attended and multi-head attention layer in STE, we set the number of heads to 128. During training, the hyper-parameter λ used to balance the preferences between the two decoders is set to 0.6, and the epochs of training are 30 and 16 for MSVD and MSR-VTT, respectively. Meanwhile, we utilize Adam optimizer [48] with a batch size of 32 and the learning rate of 1e-4 and 3e-5 for MSVD and MSR-VTT, respectively. In order to generate a better caption, we leverage the beam search with a size of 5, and the dropout rate is 0.1. All experiments were run on the RTX3060 GPUs.

4.2 Performance comparison

The performance of our proposed BTKG will be compared with the following state-of-the-art methods: RecNet [40], PickNet [49], NACF [27], STAT [46], DRPN [50], ORG-TRL [51], SGN [52], O2NA [25], MGRMP [53], GSB + CoSB [54], TVRD+OAG [55]. Unless otherwise stated, we report the results from the original papers directly. The test results of MSVD and MSR-VTT datasets are shown in Table 1, BTKG is superior to most state-of-the-art methods on both datasets.

Notably, our BTKG achieves significant improvement on CIDEr-D, which is specially designed for captioning tasks, e.g., a relative improvement of 6.1% on MSVD compared with TVRD+OAG while 6.9% on MSR-VTT compared with NACF. However, our model does not perform best on BLEU because BLEU is specially designed for machine translation and word level-based matching. Sulem et al. [56] find low or no correlation between BLEU and the grammaticality and meaning preservation parameters where sentence splitting is involved, and the research of Novicoca et al. [47] shows that BLEU is not well consistent with human judgment, whereas CIDEr-D is more in line with the writing habits of human beings [46]. Therefore, it is difficult for CIDEr-D and BLEU to improve the same model significantly, e.g., the BLEU4 of DRPN and ORG-TRL is the highest in MSVD and MSR-VTT, respectively, but the CIDEr-D is not the highest. ROUGE and METEOR are improvements of BLEU, where ROUGE focuses on the recall rate rather than the accuracy rate, and METEOR considers the stem of synonyms and comparison words (so that "running" and "run" are regarded as matching) and is explicitly used to compare sentences rather than corpus. Because our BTKG generates elegant sentences by the Extended and Multi-head Attention, which fully captures the fine-grained spatio-temporal features of the video, it performs well in these two scores.

To sum up, the excellent performance on METEOR, ROUGE, and CIDEr-D indicates that our BTKG not only obtains more crucial objects and the relational knowledge between the objects from videos through the object detection and the knowledge graph, respectively, but also maintains human language habits due to the consideration of the context in the generation of captions by a bidirectional decoder. The advantage will also be discussed through visualized examples in Section 5.2.

5 Analysis

In this section, we analyze the benchmark MSR-VTT dataset from different aspects to better understand our approach. Specifically, we first go through the quantitative analysis and introduce the case study.

5.1 Quantitative analysis

We first research the feasibility of connecting a bidirectional decoder and an encoder that integrates fine-grained spatio-temporal features, objects, and relationships between the objects in the video by conducting the ablation study. Then, we will discuss five comparative experiments as follows:

- Effect of the number of heads of the extended and multi-head attention layer;
- Effect of the number of heads used to encode the objects and relationships, respectively;
- Effect of the positional encoding on objects and relationships encoders, respectively;
- Effect of the hyper-parameter λ;

Table 1 Experimental results on msvd and msr-vtt datasets

Methods	Year	Features	RS	Dataset: MSVD [36]				Dataset: MSR-VTT [37]			
				B@4	M	R	C	B@4	M	R	C
RecNet [40]	2018	IV4	✗	52.3	34.1	69.8	80.3	39.1	26.6	59.3	42.7
PickNet [49]	2018	R152	✗	52.3	33.3	69.6	76.5	41.3	27.7	59.8	44.1
NACF [27]	2019	R101+3DR	✗	55.6	36.2	–	96.3	42.0	28.7	–	51.4
STAT [46]	2019	R152+C	✗	52.0	33.3	–	73.8	39.3	27.1	–	43.9
DRPN [50]	2020	IV4	✗	57.3	34.3	72.0	86.4	39.5	27.7	61.0	49.2
ORG-TRL [51]	2020	IRV2+C	✗	54.3	36.4	73.9	95.2	43.6	28.8	62.1	50.9
SGN [52]	2021	R101+3DR	✗	52.8	35.5	72.9	94.3	40.8	28.3	60.8	49.5
O2NA [25]	2021	R101+3DR	✗	55.4	37.4	74.5	96.4	41.6	28.5	62.4	51.1
MGRMP [53]	2021	IRV2+3DR	✗	55.8	36.9	74.5	98.5	41.7	28.9	62.1	51.4
GSB+CoSB [54]	2022	R101+3DR	✗	50.7	35.3	72.1	97.8	41.4	27.8	61.0	46.5
TVRD+OAG [55]	2022	IRV2+C+F	✓	50.5	34.5	71.7	84.3	43.0	28.7	62.2	51.8
BTKG (Ours)	2022	IRV2+I+M	✓	55.7	38.3	74.7	104.5	42.8	30.0	62.4	55.4

In this paper, the red- and the blue- colored numbers indicate that the results of all methods are the best and suboptimal, respectively. R, B@4, M, and C refer to Rouge-L, BLEU4, Meteor, and CIDEr-D, respectively. IV4, R152, R101, IRV2, 3DR, I, C, F and M denote Inception-V4, ResNet-152, ResNet-101, InceptionResNetV2, 3D ResNeXt-101, I3D, C3D, Faster-RCNN and Mask-RCNN, respectively. And the relationship (RS) column denotes whether this method uses relationships to enhance performance

– Effect of the extended and multi-head attention on ORE.

5.1.1 Ablation study

For evaluating the effectiveness of our proposed BTKG, Table 2 shows that when we directly connect KGVideo and BiTransformer, the performance of KGVideo + BiTransformer is the worst, indicating that they harm each other rather than jointly improve the overall performance, e.g., KGVideo amplifies the information leakage of BiTransformer, resulting in a worse result after connection. However, it is observed that our BTKG exceeds KGVideo and BiTransformer in most metrics, indicating that our model successfully utilizes the objects and the relationships in the video and generates more accurate captions through a bidirectional decoder.

5.1.2 Effect of the number of heads of the extended and multi-head attention

In the Spatio-Temporal Encoder, we capture the closer correlations between smaller video clips than frames through the extended and multi-head attention layer. As shown in Table 3, when increasing the number of heads, the encoder can capture adequate relevance between the clips, and the performance of our model will also be significantly improved. However, if the number of heads increases to 160, the model is prone to overfitting, and the performance declines, so we finally adopted 128 heads.

5.1.3 Effect of the number of heads on the encoders of objects and relationships

There are relatively more important objects and specific correlations between the objects in the video, whereas the relationships between the objects are entirely independent of each other. Therefore, as shown in Table 4, our model achieves the best performance when we employ the attention layer of 10-heads (NHEO = 10) to capture the correlations between the objects and do not utilize the attention layer (NHER = 0) for the relationships. Secondly, when we employ the attention layer of the heads (NHEO $<$ 10) to capture the correlations between the objects or leverage the attention layer (NHER $>$ 0) to obtain the non-existent associations between the relationships, both of these improvements can reduce the performance of our model.

5.1.4 Effect of the positional encoding on objects and relationships encoders

The objects in the video and the relationships between the objects are composed of a series of isolated entities, so there is no a sequential connection in them. The positional encoding is not crucial in their encoding. As shown in Table 5, when we add positional encoding to the

Table 2 Ablation study of BTKG performed on the MSR-VTT dataset

Method	B@4	M	R	C
BiTransformer [8]	43.3	28.3	62.0	52.8
KGVideo [7]	42.3	29.6	61.8	53.8
BiTransformer + KGVideo	41.8	28.2	61.4	51.2
BTKG (Ours)	42.8	30.0	62.4	55.4

Table 3 Effect of the number of heads of the extended and multi-head attention layer

Number of Heads	B@4	M	R	C
10	42.6	29.8	62.5	54.9
32	43.0	29.9	62.5	55.0
64	42.4	30.0	62.2	55.0
128	42.8	30.0	62.4	55.4
160	42.6	29.8	62.2	54.7

encoders of the objects or the relationships, the performance of our model can be reduced. When both do not employ positional encoding, the result is the best.

5.1.5 Effect of hyper-parameter λ

We introduce the hyper-parameter λ to balance the preferences of the forward and backward decoders on the training (see (13)). In Table 6, we leverage different λ training BTKG on the MSR-VTT to determine the value of λ. When $\lambda = 1$, the backward decoder does not participate in the training, and the performance of BTKG is the worst. When $0.5 \leq \lambda < 1$, the four metrics all have a particular improvement compared with $\lambda = 1$, i.e., the backward decoder affects the generation of captions in the forward decoder. When we set $\lambda = 0.6$, a stable balance is achieved between the two decoders, resulting in the best four metrics.

5.1.6 Effect of the extended and multi-head attention on ORE

In the ORE of Fig. 2, our BTKG enhances the representation of video features through the objects and the relationships. If we continue to employ the extended multi-head attention for encoding the fine-grained spatio-temporal features, it can increase the redundancy after a modal fusion of the video and further weaken the feature representation. As shown in Table 7, the CIDEr-D decreases by 2.8%. Therefore, in future work, we intend to propose a better method for fusing fine-grained spatio-temporal features, objects, and relationships in the video. It can take advantage of the complementarity and eliminate the redundancy between multiple modal features, rather than superimposing four feature encodings by the element-wise addition (see (11)).

5.2 Case study

Although we can evaluate our model and summarize it through the evaluation mechanism in Section 4.2, the scores can not directly reflect the performance of the sentences generated

Table 4 Effect of the number of heads used to encodes the objects (NHEO) and relationships (NHER)

NHEO	NHER	B@4	M	R	C
10	10	40.9	29.8	62.0	53.5
10	1	42.5	29.8	61.9	55.0
10	0	42.8	30.0	62.4	55.4
1	0*	42.2	29.6	61.8	53.7
0	0	41.6	29.8	62.0	54.5

* NHER = 0 indicates that the attention layer is not used

Table 5 Effect of the positional encoding (PE) on objects and relationships encoders

Objects +PE*	Relationships +PE	B@4	M	R	C
✓	✓	42.8	29.9	62.1	54.9
✓	✗	42.6	29.8	62.3	54.9
✗	✓	42.8	29.9	62.3	55.0
✗	✗	42.8	30.0	62.4	55.4

∗ +PE represents adding positional encoding

Table 6 Effect of the hyper-parameter λ

Hyper-parameter λ	B@4	M	R	C
0.5	42.2	29.7	62.1	54.7
0.6	42.8	30.0	62.4	55.4
0.7	42.3	29.9	62.2	54.7
0.8	42.6	29.9	62.3	54.9
0.9	42.8	29.7	62.3	54.7
1.0	41.5	30.0	61.7	54.4

Table 7 Effect of the extended and multi-head attention (EMHA) on ORE

ORE +EMHA	B@4	M	R	C
✓	42.0	29.5	61.8	53.9
✗	42.8	30.0	62.4	55.4

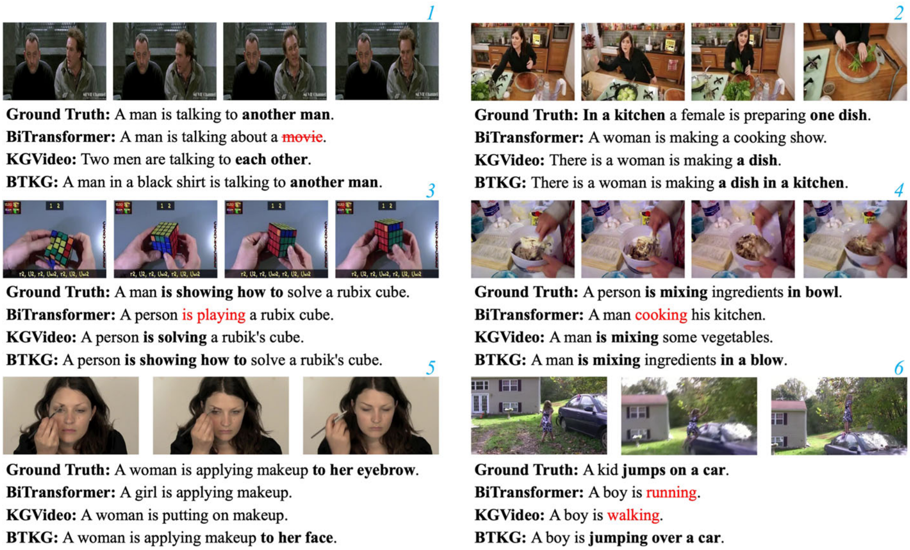

Fig. 3 Visualized examples of captions generated by BiTransformer [8], KGVideo [7], our proposed BTKG, and human annotation (Ground Truth), of which the first three and the last three are from MSR-VTT and MSVD, respectively

by BTKG. Therefore, in Fig. 3, we list a few visualized examples of captions generated by BiTransformer [8], KGVideo [7], BTKG, and human annotation (Ground Truth) to analyze and verify the summary made in Performance Comparison.

In *No.1* and *No.2* videos of Fig. 3, both KGVideo and BTKG acquire the objects in the predicate in the video(e.g., 'each other', 'another man', 'a dish') through object detection. Whereas, BiTransformer only obtains the object 'person' in the subject , and even used the wrong word 'moive' in the *No.1* video. In addition, the model can describe the relationships between objects more accurately thank to the fusion of a knowledge graph model. For example, in *No.3* and *No.4* videos, KGVideo and BTKG utilize the two relational words 'solving' and 'mixing', and even in the *No.3* video, BTKG uses 'is showing how to solve' through the formula to solve the rubix cube in the video, whereas BiTransformer uses only two abstract verbs, 'playing' and 'cooking', to summarize the relationships between objects. Lastly, our BTKG employs an encoder that combines fine-grained spatio-temporal features, objects, and the relationships between objects in the video while using the bidirectional decoder. The generated sentences are more in line with human language habits, and scene words ('in a kitchen', 'in a bowl', 'to her face') are added to embellish the sentences when describing the video. Even in *No.6*, only our BTKG recognizes the behavior of 'jumping over a car', while the other two models utilize the abstract and inaccurate verbs 'running' and 'walking'.

Because our model generates more scene words, the captions generated by BTKG are more in line with human language habits. Even in the *No.1*, a modifier 'in a black shirt' is reasonable, but it does not appear in Ground Truth. It further explains why our BTKG has a higher CIDEr-D and a smaller increase on BLEU in the Performance Comparison.

6 Conclusion

In this paper, we propose BTKG, which simultaneously enhances both the encoder and decoder of the Transformer. In the encoder, we not only utilize common video features such as image, motion, and object features, making fine-grained improvements tailored to their respective attributes during encoding, but also introduce a knowledge graph that extracts relationship feature between objects within the video. Since the Transformer decoder is autoregressive, where the generation of each word depends on the preceding outputs, in the decoder, we employ a bidirectional decoder to consider context while generating each word. In addition, due to the problem of information leakage in the bidirectional decoder, the incorporation of object and relationship features into the encoder exacerbates this problem, leading to a decrease in model performance. Therefore, we mitigate the leakage problem by generating pseudo reverse captions and employing dual encoders. Experiments on the MSVD and MSR-VTT datasets demonstrate that our proposed BTKG achieves state-of-the-art performance in significant metrics. Moreover, the sentences generated by BTKG contain scene words and modifiers, that are more in line with human language habits.

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Data Availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of Interest The authors declare that they have no conflict of interest.

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